

Observation and Analysis of a Multiple Lightning Strike Based on Dynamic Vision

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Abstract—To capture lightning more conveniently and efficiently, this paper employs dynamic vision sensor technology to collect lightning data. Compared to high-speed cameras, dynamic vision sensors have the advantages of smaller data volume and lower cost; compared to conventional cameras used in power grids, dynamic vision sensors offer higher temporal resolution. This study employs dynamic vision technology to observe and analyze lightning strikes. This article develops a set of event-based image reconstruction and clustering algorithms to analyze a multiple cloud-to-ground lightning strike that occurred on May 6, 2024, restoring the lightning process and extracting the lightning channel. Firstly, the event data is reconstructed into lightning images, revealing that this cloud-to-ground lightning event includes one leader discharge and three return strokes, with an average time interval of 60ms between each discharge. Additionally, the K-Means clustering algorithm is employed to extract the lightning channel portion from the pixel image. This research utilizes dynamic vision sensor technology to study the development and channel extraction of multiple lightning strikes, providing a new method and effective approach that can serve as a powerful complement to traditional lightning observation techniques.

Keywords—dynamic vision technology, lightning optical observation, multiple lightning strike, clustering algorithm, lightning channel

I. INTRODUCTION

Lightning is a momentary discharge event occurring in the atmosphere, characterized by long distances, high currents, strong radiation, and shock waves. Its immense energy and electromagnetic effects have a significant impact on human society and the natural environment [1].

According to the TB 549 report published by the CIGRE, over 80% of cloud-to-ground lightning strikes are multiple strikes, which severely affect the stable operation of transmission lines [2]. Multiple strikes refer to the phenomenon where multiple discharges occur along the same discharge channel (or nearby channels) between the cloud and the ground or between clouds during a single thunderstorm. These multiple strikes have short time intervals and high energy density, posing challenges for lightning observation and data analysis. The waveforms of multiple strikes entering the power system are more complex than those of single discharges; the randomly superimposed continuous multiple pulse over-voltages often have higher amplitudes and longer durations, causing severe load impacts on electrical equipment [3]. Therefore, studying the observation of multiple lightning discharge processes not only aids in a deeper analysis of their complex characteristics but also holds significant importance for lightning protection design and equipment safeguarding.

Currently, the main equipment used for lightning observation includes high-speed cameras and lightning channel imagers. However, high-speed cameras require substantial data processing and storage capacity, and their limited recording time makes it inconvenient to capture natural lightning, resulting in low data acquisition efficiency [4], [5], [6], [7]. Another method involves using lightning channel imagers to record images of lightning channels; however, these imaging devices have relatively low frame rates, with a maximum of only 75 frames per second (fps), making them unsuitable for capturing the rapid development of multiple strikes [8], [9], [10]. This study employs a Dynamic Vision Sensor (DVS) that captures changes in light energy based on an energy "differential" approach, utilizing an event-driven mechanism to output an asynchronous sparse data stream. Compared to traditional frame cameras limited by frame rates and expensive, bulky, high-power-consuming high-speed cameras, DVS offers advantages such as low latency, high temporal resolution, high dynamic range, low bandwidth, and low power consumption.

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Therefore, this study utilizes DVS to record a complete cloud-to-ground lightning event in Guangzhou. Through a series of algorithmic processes applied to the event stream, the number of discharges and the starting time of each discharge are obtained, along with the reconstruction of the lightning discharge channel's position and morphology.

II. METHODS

A. Observational Equipment

In this study, DVS technology is employed to assist in the optical observation of lightning activity. DVS capture variations of light energy in the form of an "energy difference" to perceive scenes. Each pixel operates independently and asynchronously. DVS are noted for their sensitivity to brightness changes, rapid response speed, high temporal resolution up to 10^6 fps, wide dynamic range (> 120 dB), and small data volume (approximately 1/100th that of high-speed cameras) [11]. These features facilitate the detection of fastmoving objects, such as bullets, dynamic obstacle avoidance, and target tracking [12], [13], [14].

Each pixel in the DVS sensor array operates asynchronously. A new event is triggered if the brightness change at a certain pixel meets the following condition:

$$\log(I(t, x, y)) - \log(I(t - \Delta t, x, y)) \geq C \quad (1)$$

DVS stores events in the format of (t, x, y, p) , where $t - \Delta t$ represents the timestamp of the last event triggered for that pixel, (x, y) indicates the spatial coordinates of the pixel, C denotes the event triggering threshold (Contrast Threshold) of the DVS, and $I(t, x, y)$ represents the absolute grayscale value of the pixel at time t . When the brightness change exceeds the threshold, a new event is generated. The variable p indicates the polarity of the light intensity change, where $p=1$ signifies an increase in light intensity (output ON event), and $p=-1$ signifies a decrease in light intensity (output OFF event).

This article employs DVS technology for the optical observation of lightning. During the thunderstorm season in the southern region, when lightning alerts are issued, a DVS, as shown in Fig. 1, is fixed on a building in an area with a wide field of view to collect signals of light intensity changes directed at the sky. The dynamic visual data is saved in real-time as .aedat files on a computer for subsequent processing. The content of this article is based on the analysis of data from a multiple lightning event. The DVS used in this study is the Davis 346 produced by Inivation (acquired by SynSense), with a time resolution on the order of microseconds and a spatial resolution of 346×260 . The event output features a dynamic range of up to 120 dB and a latency of less than 1 μ s.

TABLE I. COMPARISON OF THE DVS AND THE FRAME CAMERA.

Camera	Data volume (G/h)	Power consumption (W)
Dynamic vision sensor	1	0.457
High-speed industrial camera	287.98	≥ 1.63

The dimensions of the device are $40 \times 60 \times 25$ mm, and it weighs 100 g. Additionally, we compared the power consumption and data volume of the DVS and frame camera (as shown in Table I). It was found that the data volume of the DVS is significantly smaller than that of the high-speed industrial camera.

III. DATA AND PROCESSING

In the 2024 thunderstorm season, dynamic visual sensors were employed for lightning observation, utilizing the Davis 346 for lightning capture in Guangzhou. This study focuses on the observation and analysis of multiple lightning strikes based on dynamic visual technology, with the analysis derived from data collected during a cloud-to-ground lightning event that occurred on the evening of May 6, 2024. The data collected by the Davis 346 includes event streams and frame images, with a data size of 15MB over a duration of 10 seconds.

A. The Process of Cloud-to-ground Lightning

The multi-strike cloud-to-ground lightning occurred on May 6, 2024, at 21:09. As shown in Fig. 2-Fig. 5, this lightning process includes a leader discharge followed by three return strokes (Green represents ON events, while red represents OFF events). The time intervals between each discharge were 44,700 μ s, 100,300 μ s, and 35,000 μ s, with an average time interval of 60ms.



Fig. 1. Optical observation experiment site and dynamic vision sensor.

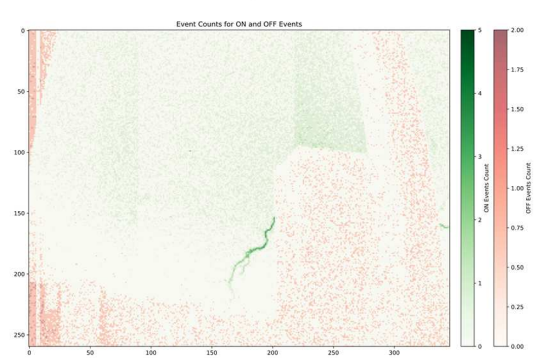


Fig. 2. The reconstructed image of leader discharge (3540000-3565000 μ s).

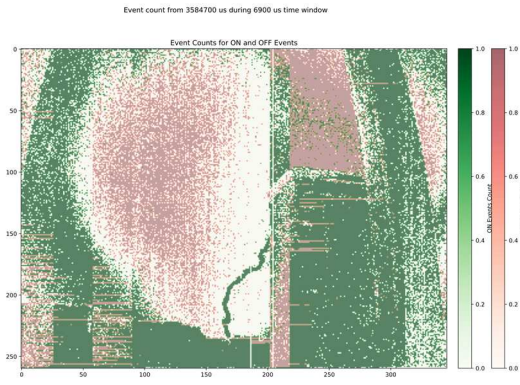


Fig. 3. The reconstructed image of the first return stroke (3584700-3585000 μ s).

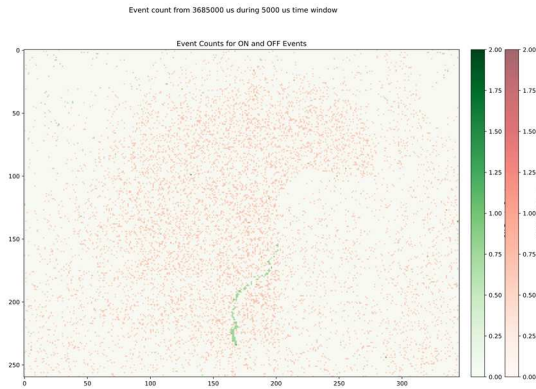


Fig. 4. The reconstructed image of the second return stroke (3685000-3693000 μ s).

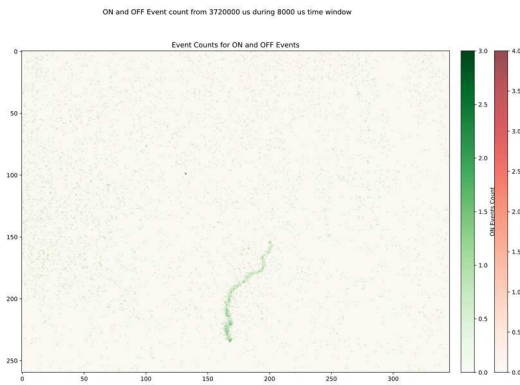


Fig. 5. The reconstructed image of the third return stroke (3720000-3730000 μ s).

Due to the presence of a large area of buildings (such as structures and ground) aside from the sky in the recorded footage, a majority of the pixels captured a significant amount of light intensity variation in a short period during each discharge. This resulted in the simultaneous occurrence of many other events in the footage, in addition to the lightning channel. Therefore, to accurately extract the lightning channel, it is necessary to design an algorithm to segment the main objects in the footage (including the lightning channel, sky, buildings, and ground) to effectively isolate the lightning channel area.

B. Extract Lightning Channel

The start and end time of this lightning event is 3540000-4050000 μ s. Based on the sequence and polarity differences of the emitted events, the objects in the scene are divided into

three regions: (1) lightning channel, (2) background sky, and (3) buildings and ground. During a discharge process, the lightning channel region emits ON/OFF events first, followed by the other regions (background sky and buildings and ground) which begin to emit ON/OFF events. At the same moment, different regions may emit events of different polarities. According to the alternating characteristics of the ON/OFF events in each region, the time period of this lightning event is divided into 18 segments (as shown in Fig. 6), and the total number of ON events and OFF events for all pixel points in each time period is counted. This analysis provides important temporal reference points for the subsequent division of regions.

To delineate the pixel points of the lightning channel area, we have designed a clustering algorithm process, as shown in Fig. 7. This process includes the following steps:

(1) Data Slicing. The data is sliced based on 18 time segments and two polarities (ON/OFF), with each pixel point corresponding to a 36-dimensional matrix (18 time segments $\times$ 2 polarities). The entire dataset contains $346 \times 260 = 89,960$ 36-dimensional matrices.

(2) Feature Expansion. To enhance spatial continuity, the x and y coordinates of each pixel point are added to the feature matrix, thereby expanding the feature space. Now, each pixel corresponds to a 38-dimensional matrix.

(3) Feature Normalization. StandardScaler is used for normalizing the data, ensuring that all features are within the same scale range, which improves data comparability and optimizes the performance of the subsequent clustering algorithm.

(4) Feature Weighting. Higher weights are assigned to the feature dimensions corresponding to the time slices related to lightning occurrences, with a weighting value set at 2. Simultaneously, the x and y coordinates are also weighted, with a weighting value of 1.3, to further enhance spatial

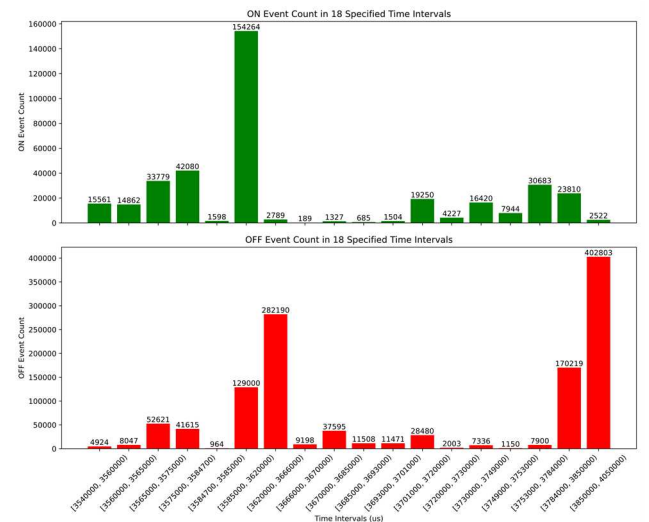


Fig. 6. Event counts across 18 times segments.

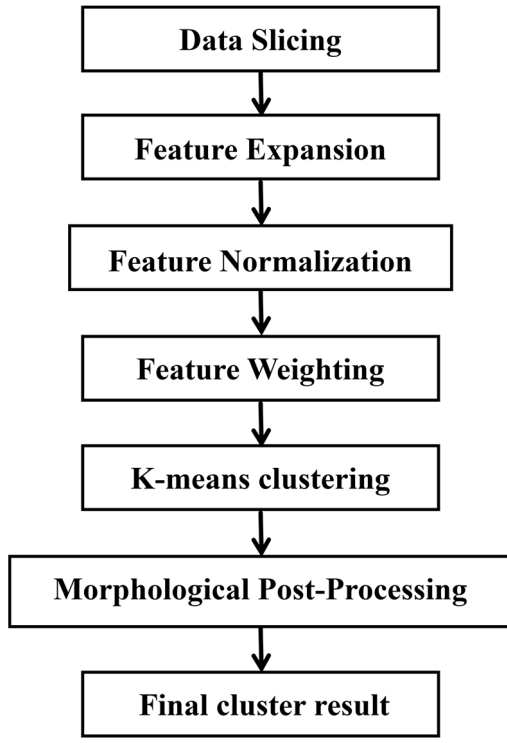


Fig. 7. Lightning clustering algorithm process.

continuity. The weights for the remaining dimensions are all set at 1.

(5) K-Means Clustering. The K-Means clustering algorithm is employed for clustering analysis. Initially, three points are randomly selected as the initial cluster centers, and the distance from each data point to the three cluster centers is calculated, assigning each point to the nearest cluster. Subsequently, the mean of all data points in each cluster is computed and designated as the new cluster center. This process is repeated until the algorithm converges.

(6) Morphological Post-Processing. Basic operations of mathematical morphology (dilation and erosion) are applied to post-process the clustering results, smoothing object boundaries and eliminating small noise. The final clustering results, as shown in Fig. 8, indicate that the clustering value for building and ground areas is 0, the background sky has a

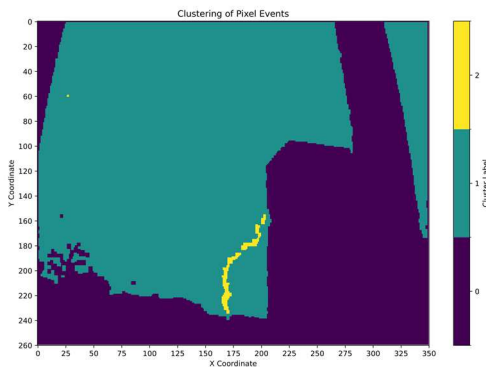


Fig. 8. Clustering result graph.

clustering value of 1, and the lightning channel has a clustering value of 2.

This algorithm process leverages the temporal and spatial features of the events to successfully achieve effective classification and reconstruction of the lightning channel.

IV. CONCLUSION

This paper applies DVS technology to the study of lightning observation, analyzing the evolution of multiple lightning strikes based on dynamic visual data. The research focuses on a specific instance of multiple lightning phenomena, including the following aspects:

Firstly, for a cloud-to-ground lightning process, the data is segmented by utilizing the image reconstruction results of the events. The analysis indicates that this cloud-to-ground lightning event consists of one preliminary discharge and three return strokes, with an average time interval of 60ms between the two discharges.

Secondly, by conducting a regional analysis of the cloud-to-ground lightning, it is divided into 18 times segments, and the K-Means clustering algorithm is employed to achieve regional partitioning, thereby accurately extracting the pixels positions of the lightning channels.

This study provides new perspectives and methodologies for the observation and analysis of multiple lightning strikes, overcoming the limitations of traditional frame cameras in capturing lightning and further expanding the application scope of DVS technology. Notably, DVS technology offers advantages such as small size, low data volume, and low power consumption, enabling widespread deployment and multi-site arrangements, which facilitates the efficient accumulation of natural lightning data. Additionally, DVS technology can be integrated with Spiking Neural Networks (SNN) and brain-like computing systems to construct a unified sensing, storage, and computing system, thereby meeting the demand for real-time online monitoring of lightning. In the future, DVS technology holds broad application prospects in fields such as meteorology, aerospace, the power industry, and disaster monitoring.

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